

RESEARCH THESIS

VANTA

**Virtual Aperture from Nonlinear Transfer
Artifacts**

Receiver-State Spectroscopy, Nonlinear Transfer
Characterization,
and Partial Physical Validation with an RTL-SDR Blog V4
Receiver

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Abstract

This thesis develops and experimentally evaluates VANTA (Virtual Aperture from Nonlinear Transfer Artifacts), a receiver-centric RF sensing concept in which nonlinear transfer artifacts are treated not only as impairments but also as candidate auxiliary observables. The central premise is that a receiver whose transfer law depends on an internal state, such as tuner gain, may expose state-dependent spectral structure that is absent from an intentionally restricted linear observation. In the constructive model used here, third-order mixing products at $g_1 = 2f_1 - f_2$ and $g_2 = 2f_2 - f_1$ permit algebraic reconstruction of two parent frequencies through $f_1 = (2g_1 + g_2)/3$ and $f_2 = (g_1 + 2g_2)/3$. The work therefore separates two questions that are often conflated: whether receiver nonlinearity can be physically observed and characterized, and whether specific nonlinear products can be causally linked to hidden parent signals strongly enough to support inverse reconstruction.

The investigation proceeded in two layers. Layer 1 used a controlled discrete-time nonlinear model to establish a constructive proof of concept. The simulation produced the expected second- and third-order spectral components, reproduced state-dependent IM3 behavior, recovered the two parent frequencies exactly on the chosen frequency grid, recovered amplitudes with a maximum relative error of approximately $1.7 \times 10^{-6}\%$, increased the selected observation Jacobian rank from zero to two, and passed all seven predefined simulation gates. Monte Carlo testing retained full recovery for input SNR values of -5 dB and above in the recorded campaign, while the lowest tested SNR point produced a reduced recovery rate. This simulation is interpreted as a constructive identifiability example rather than a general theorem.

Layer 2 moved to a real RTL-SDR Blog V4 receiver and an outdoor ambient FM environment. After several acquisition and measurement-design iterations, the final campaign used direct asynchronous IQ capture at 2.048 MS/s, 250,000 complex samples per capture, 28 overlapping survey centers from 88 to 108 MHz, and five manual tuner-gain states. The final stitched survey achieved 100% geometric coverage at every gain state and passed non-stale-IQ validation. A real RF target at 95.097007 MHz exhibited a prominence of 34.339 dB. A safe-region gain model yielded $R^2 = 0.974319$ and gain-response correlation $\rho = 0.987076$. The measured response departed strongly from the safe extrapolation beginning at a tuner-gain command of 25.4 dB, with a compression-depth metric of 22.719 dB, maximum stress-state clipping fraction of 0.686344, and median two-repeat power difference of 0.074 dB. A robust frequency-dependent residual analysis produced a scale estimate of 5.605 dB, an adaptive threshold of 25.221 dB, and two spectral bins above threshold, thereby establishing

nonlinear spectral redistribution under the adopted gates. LO-coordinate validation of the 95.097007 MHz target achieved an RF tracking standard deviation of 0.244 kHz, maximum baseband-coordinate error of 3.997 kHz, and median validation SNR of 7.523 dB.

The final ambient IM3 stage tested four candidate parent pairs. Although product-only algebraic reconstruction errors ranged from 31.848 to 76.554 kHz, the candidate products did not satisfy the physical validation requirements: their SNR values were negative, and several candidates exhibited excessive baseband tracking error or RF-coordinate variance. Consequently, the receiver-nonlinearity and measurement-integrity gates passed, while the ambient intermodulation-graph and product-only parent-reconstruction gates failed. The final physical conclusion is therefore deliberately limited: receiver-state nonlinear compression and frequency-dependent spectral redistribution were experimentally established, but causal ambient IM3 parent reconstruction was not established. This distinction is a central result of the thesis because it prevents algebraic coincidence from being misreported as physical nonlinear inversion.

Keywords: software-defined radio; nonlinear receiver; intermodulation distortion; receiver-state spectroscopy; inverse problems; RTL-SDR Blog V4; RF signal processing; dynamic range; spectral redistribution.

Executive Summary of Findings

The final evidence is best summarized as a hierarchy rather than as a single binary claim. First, the measurement platform itself was validated: the final campaign completed the full 88–108 MHz survey with 100% stitched coverage, finite and changing IQ fingerprints, repeatable target-band measurements, and LO-coordinate confirmation of a real RF signal. Second, receiver nonlinearity was physically demonstrated. Strong signals that followed a nearly linear gain trend at low and moderate tuner-gain states ceased to scale linearly as the receiver was driven into high-gain states; clipping rose sharply and the measured target-band power saturated relative to the safe-region extrapolation. Third, nonlinear behavior was not purely common-mode. After common compression was removed, two frequency bins exceeded a conservative robust threshold for stress-state spectral redistribution. Fourth, the stronger VANTA claim – that ambient nonlinear products could be used as a physically verified graph from which parent frequencies could be reconstructed – was not established. All four final candidate pairs failed at least one physical product gate, principally because the candidate product SNRs were below zero dB and several baseband-coordinate errors were too large.

This outcome is scientifically useful. A less disciplined analysis could have reported the small algebraic reconstruction errors of some candidate pairs as evidence of hidden-source recovery. The present work instead requires the measured products themselves to behave as real RF quantities under LO shifts and to possess sufficient SNR before algebraic inversion is accepted. Under those gates, the ambient FM environment did not provide an acceptable pair. The final status is therefore `RECEIVER_NONLINEARITY_PASS_AMBIENT_IM3_NOT_ESTABLISHED`. That status is not a software failure; it is the experimental answer produced by the measurement protocol.

Nomenclature

Symbol / Term	Definition
A_1, A_2	Parent-signal amplitudes in the constructive nonlinear model.
f_1, f_2	Parent RF or baseband frequencies.
g_1, g_2	Third-order difference products $2f_1 - f_2$ and $2f_2 - f_1$.
G	Receiver tuner-gain command in dB.
$P(f, G)$	Measured spectral power quantity at RF frequency f and receiver state G .
$R(f, G)$	Stress-state residual relative to a safe-region linear gain model.
$S(f, G)$	Frequency-dependent residual after removal of common-mode stress response.
σ_{MAD}	Robust scale estimate obtained from the median absolute deviation.
J	Jacobian of an observation operator with respect to active scene parameters.
$\mathcal{F}_\theta$	State-dependent receiver observation operator.
θ	Receiver state, including tuner gain, local oscillator, and other controllable settings.
ADC	Analog-to-digital converter.
AGC	Automatic gain control.
BB	Baseband.
CW	Continuous wave.
DFT / FFT	Discrete Fourier transform / fast Fourier transform.
IM2 / IM3	Second- / third-order intermodulation products.
LO	Local oscillator.
MAD	Median absolute deviation.
PSD	Power spectral density.
RF	Radio frequency.
SDR	Software-defined radio.
SFDR	Spurious-free dynamic range.
VANTA	Virtual Aperture from Nonlinear Transfer Artifacts.

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Introduction

1.1 Motivation

Conventional receiver design treats nonlinear distortion as an impairment. The usual engineering objective is to maintain operation inside a region where the desired signal follows a sufficiently linear transfer law, while harmonics, compression, intermodulation products, quantization artifacts, and spurious responses remain below acceptable limits. This view is fully justified for communication receivers, measurement instruments, and RF front ends whose primary task is faithful signal reproduction. Standard discussions of receiver dynamic range therefore place noise at the lower performance boundary and compression or intermodulation at the upper boundary [1, 3, 7]. Automatic gain control is similarly used to prevent internal stages from being driven outside their intended linear region [5, 7].

The same physical facts can, however, be viewed from a different measurement perspective. A deterministic distortion product is not random merely because it is undesirable. If its frequency, amplitude, or phase depends reproducibly on otherwise unobserved input structure, then it is a transformation of information. The central question of VANTA is therefore not whether receiver nonlinearity is beneficial in the conventional communication sense; it is whether some reproducible nonlinear transfer artifacts can be treated as auxiliary observables in an inverse problem.

This thesis develops that question in a deliberately conservative manner. The receiver is not assumed to be an ideal polynomial device. Nor is every spur interpreted as useful. Instead, the work begins with a controlled constructive model in which the mapping from parent tones to nonlinear products is analytically known, then progressively introduces real receiver states, ambient broadcast signals, clipping, receiver-coordinate artifacts, nonstationarity, and hardware acquisition constraints. Each stage is assigned explicit gates. A claim is accepted only when the corresponding gates are satisfied.

1.2 Research Problem

Let an unknown sparse RF scene be represented by a parameter vector $\mathbf{x}$. A conventional linearized receiver state can be idealized as

$$\mathbf{y}_{\text{lin}} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (1.1)$$

where $\mathbf{H}$ includes the receive chain and observation mask, and $\mathbf{n}$ denotes noise. If the observation operator suppresses or excludes part of the support of $\mathbf{x}$, the corresponding parameters may not be identifiable from eq. (1.1) alone.

A real receiver need not remain linear as its internal state changes. The more general observation is written as

$$\mathbf{y}_k = \mathcal{F}_{\theta_k}(\mathbf{x}) + \mathbf{n}_k, \quad (1.2)$$

where θ_k denotes the k th receiver state. The proposed virtual aperture is not a physical antenna aperture; it is the enlarged set of measurement constraints formed by evaluating the same scene through multiple state-dependent transfer operators. In the most optimistic case,

$$\mathcal{I}(\mathbf{x}; \mathbf{y}_1, \dots, \mathbf{y}_K) > \mathcal{I}(\mathbf{x}; \mathbf{y}_{\text{lin}}), \quad (1.3)$$

where $\mathcal{I}$ denotes mutual information. Equation (1.3) is a research objective rather than a result established by the present hardware experiment.

The practical problem addressed here is narrower and testable: can manual receiver-state changes reveal reproducible nonlinear spectral behavior, and can any resulting ambient IM3 products be validated strongly enough to support parent-frequency reconstruction?

1.3 Research Questions

The investigation was organized around four research questions:

- RQ1:** Does the RTL-SDR receiver exhibit a repeatable state-dependent response that departs from a safe-region linear gain model under strong ambient FM excitation?
- RQ2:** After common compression is removed, does the stress-state spectrum contain frequency-dependent residual structure that is statistically large relative to the robust residual distribution?
- RQ3:** Can a selected target be shown to occupy a fixed RF coordinate when the local oscillator is shifted, thereby distinguishing a real RF feature from a receiver-fixed baseband artifact?

RQ4: Do ambient candidate products satisfy the physical requirements of an IM3 graph and support product-only parent-frequency reconstruction?

These questions intentionally separate the existence of receiver nonlinearity from the stronger inverse-reconstruction claim. The final experiment answers RQ1–RQ3 positively under the adopted gates and RQ4 negatively.

1.4 Principal Contributions

The work makes five practical contributions. First, it formulates receiver-state spectroscopy as a measurement procedure in which the receiver state is deliberately swept and the resulting transfer surface is analyzed rather than discarded. Second, it develops a safe-versus-stress model that distinguishes ordinary gain scaling from compression and frequency-dependent residual redistribution. Third, it establishes a robust experimental validation chain that includes coverage, non-stale IQ, repeatability, clipping, SNR, and LO-coordinate tests before nonlinear inversion is considered. Fourth, it produces real hardware evidence of receiver compression and frequency-dependent spectral redistribution on an RTL-SDR Blog V4. Fifth, it documents a negative ambient IM3 result and shows why small algebraic reconstruction error alone is insufficient evidence of physical intermodulation.

1.5 Claim Discipline

Two terminological boundaries are important throughout this thesis. The measured “compression onset” is a receiver-state metric obtained by sweeping tuner-gain commands while the external RF environment remains uncontrolled. It is *not* a calibrated 1-dB compression point, because the independent variable is not a known swept input power at the receiver reference plane. Likewise, no input or output third-order intercept point is reported. Standard IP3 measurement requires controlled two-tone excitation and calibrated power scaling [1–3]. The present experiment instead characterizes the behavior of a complete low-cost receiver under ambient excitation.

The second boundary concerns parent reconstruction. The inversion equations are exact for the assumed pair of IM3 difference frequencies, but the presence of numerically compatible spectral features does not prove that those features are generated by the candidate parents. The final ambient test therefore requires positive product SNR and stable RF/baseband behavior in addition to algebraic consistency. This decision is essential to the scientific interpretation of the final result.

Technical Background and Literature Context

2.1 Software-Defined Receivers and Discrete-Time Observation

A software-defined radio shifts a substantial portion of receiver functionality from fixed analog hardware into digital processing. The concept has been discussed for decades as a means of making radio architectures reconfigurable through software [10, 11]. In a practical low-cost SDR, however, the receive path remains a mixed-signal system: analog filtering, tuner gain, frequency conversion, ADC quantization, USB transport, and digital processing jointly determine the available data.

The discrete-time representation used in this thesis follows standard sampling and Fourier-analysis principles [5, 6]. For a continuous-time received waveform $r(t)$ sampled at interval T_s , the complex sequence is

$$r[n] = r(nT_s), \quad T_s = \frac{1}{f_s}. \quad (2.1)$$

For a finite record of N samples, a DFT-based spectrum is

$$R[k] = \sum_{n=0}^{N-1} r[n] e^{-j2\pi kn/N}. \quad (2.2)$$

The final hardware campaign used $f_s = 2.048$ MS/s and an 8192-point spectral estimator, giving a nominal FFT-bin spacing

$$\Delta f = \frac{2.048 \times 10^6}{8192} = 250 \text{ Hz}. \quad (2.3)$$

The processing used multiple short, windowed spectral segments, conceptually following the variance-reduction motivation of averaged modified periodograms [12]. Windowing is necessary because finite records create spectral leakage; the general trade-offs of DFT windows are treated comprehensively by Harris [13].

2.2 Receiver Dynamic Range, Gain Control, and Compression

Receiver dynamic range is bounded at the low end by noise and at the high end by distortion, compression, or spurious responses [1, 7]. Poisel explicitly notes that AGC is commonly used to hold receiver stages within their linear operating region [7]. In the present work AGC was avoided during the state sweep; instead, discrete manual tuner-gain commands were used as controlled receiver states.

A simple amplitude transfer law may be expanded as a power series,

$$y(t) = a_0 + a_1x(t) + a_2x^2(t) + a_3x^3(t) + \cdots, \quad (2.4)$$

which is a standard local model for weak nonlinear behavior [2, 3]. Cripps emphasizes that such truncated series are most useful in a weakly nonlinear neighborhood and become inadequate when clipping and higher-order behavior dominate [2]. That caution is directly relevant to the final gain sweep: once clipping fractions approach tens of percent, the receiver is in a strongly nonlinear region and a cubic-only description should not be interpreted as a complete physical model.

2.3 Two-Tone Intermodulation

For two real input tones,

$$x(t) = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t, \quad (2.5)$$

substitution into the cubic term of eq. (2.4) produces harmonics and cross-products. The cross term

$$3A_1^2 A_2 \cos^2 \omega_1 t \cos \omega_2 t \quad (2.6)$$

contains components at $2\omega_1 + \omega_2$ and $2\omega_1 - \omega_2$, while the reciprocal cross term produces $2\omega_2 \pm \omega_1$. The two in-band difference products are therefore

$$g_1 = 2f_1 - f_2, \quad (2.7)$$

$$g_2 = 2f_2 - f_1. \quad (2.8)$$

These are the familiar third-order two-tone intermodulation products described in RF and microwave texts [1–3]. Because they can lie close to the original tones, they are difficult to remove with ordinary filtering [1].

In the weakly nonlinear region, a first-order component grows with slope one on a dB-in/dB-out plot, while a third-order component grows with slope three [1]. This behavior

underlies conventional IP3 measurement. The present ambient experiment did not perform a calibrated two-tone input-power sweep and therefore does not claim IP3; nevertheless, the standard theory motivates the search for third-order structure.

2.4 Invertibility of the IM3 Frequency Map

The frequency relation may be written in matrix form,

$$\begin{bmatrix} g_1 \\ g_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}. \quad (2.9)$$

The determinant is three, so the mapping is invertible. Its inverse is

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}, \quad (2.10)$$

which yields

$$\hat{f}_1 = \frac{2g_1 + g_2}{3}, \quad (2.11)$$

$$\hat{f}_2 = \frac{g_1 + 2g_2}{3}. \quad (2.12)$$

Equations (2.11)–(2.12) are exact algebraic identities under the assumed product assignment. Their validity does not, by itself, establish that measured peaks are true intermodulation products. This distinction becomes decisive in the final ambient experiment.

2.5 State-Dependent Nonlinear Observation

For a complex discrete-time signal, a convenient behavioral model used in the constructive layer is

$$y[n] = c_{10}x[n] + c_{20}x^2[n] + c_{02}|x[n]|^2 + c_{21}x[n]|x[n]|^2 + w[n]. \quad (2.13)$$

The coefficient c_{21} controls a third-order in-band term. A receiver-state sweep may therefore be represented abstractly as

$$y_k[n] = \mathcal{F}_{\theta_k}\{x[n]\} + w_k[n], \quad (2.14)$$

where θ_k includes gain and any other state that modifies the transfer law. If a nonlinear artifact amplitude changes monotonically with a controlled state while a corresponding linear-only masked observation contains no support at that frequency, the artifact acts as an additional measurement feature.

2.6 Local Identifiability and Sparse Inversion

Let $\mathbf{x}_S$ contain the active parameters of a sparse scene. Local identifiability of a state-dependent observation can be examined through the Jacobian

$$\mathbf{J}_{\Theta,S}(\mathbf{x}) = \frac{\partial \mathcal{F}_{\Theta}(\mathbf{x})}{\partial \mathbf{x}_S}. \quad (2.15)$$

A necessary local condition for unique recovery of $|S|$ active parameters is

$$\text{rank } \mathbf{J}_{\Theta,S} = |S|. \quad (2.16)$$

The Layer 1 simulation deliberately constructed a masked observation for which the selected linear Jacobian had rank zero while the nonlinear observation had rank two. This is a constructive example, not a statement that arbitrary nonlinear receivers increase identifiability.

For a discretized sparse scene, the broader inverse problem can be written as

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \geq 0} \|\mathbf{y} - \mathcal{F}_{\Theta}(\mathbf{x})\|_2^2 + \lambda \|\mathbf{x}\|_1, \quad (2.17)$$

which connects the concept to sparse reconstruction and compressed sensing [17, 18]. Equation (2.17) is presented as a general formulation; the final hardware campaign used direct feature extraction and algebraic inversion rather than a large-scale nonlinear sparse optimizer.

VANTA Concept and Measurement Architecture

3.1 Virtual Aperture as an Operator Set

The word “aperture” in VANTA is used in an informational, not geometric, sense. A physical antenna array samples a field at multiple spatial locations. A receiver-state aperture instead samples the same RF environment through multiple transfer laws. If the state index is $k = 1, \dots, K$, the combined observation is

$$\mathbf{Y}_{\Theta} = \begin{bmatrix} \mathcal{F}_{\theta_1}(\mathbf{x}) \\ \mathcal{F}_{\theta_2}(\mathbf{x}) \\ \vdots \\ \mathcal{F}_{\theta_K}(\mathbf{x}) \end{bmatrix}. \quad (3.1)$$

The aim is to determine whether the stacked operator exposes additional scene-dependent structure. This idea does not imply that nonlinear distortion is universally beneficial; rather, it asks whether deterministic, state-sensitive distortion can sometimes be exploited after being characterized.

3.2 Receiver-State Spectroscopy

Receiver-state spectroscopy is defined here as the systematic measurement of a spectral quantity over both RF frequency and a controlled receiver state:

$$\mathcal{V}(f, G) = P(f, G), \quad (3.2)$$

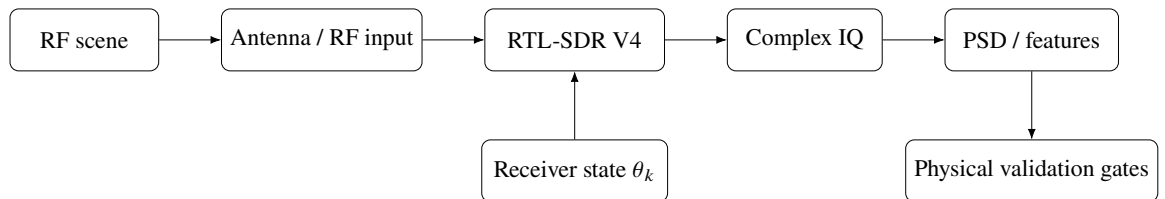


Figure 3.1: Conceptual receiver-state measurement chain used in the physical VANTA campaign.

where G is the tuner-gain command. A purely linear receiver with ideal gain control would produce approximately parallel spectral traces differing by the commanded gain. A real receiver departs from this behavior because analog stages compress, the ADC clips, noise contributions shift, filters and mixers may respond differently, and individual spectral components may redistribute energy under stress.

The final experiment did not assume that every state was equally informative. Instead, the gain states were divided into a *safe* set and a *stress* set. The safe set was used to learn a local linear response; the stress set was compared with that extrapolation. This makes the nonlinear evidence relative to the receiver's own low-stress behavior rather than to an idealized manufacturer model.

3.3 Safe-Region Linear Model

At each RF grid point f_i , the safe-state powers were approximated by

$$P(f_i, G) \approx a(f_i)G + b(f_i), \quad G \in \mathcal{G}_{\text{safe}}. \quad (3.3)$$

The coefficient of determination was calculated as

$$R^2(f_i) = 1 - \frac{\sum_k [P(f_i, G_k) - \hat{P}(f_i, G_k)]^2}{\sum_k [P(f_i, G_k) - \bar{P}(f_i)]^2}. \quad (3.4)$$

For the selected target, the same principle was applied to target-band power as a function of gain.

3.4 Stress Residual and Common-Mode Removal

For a stress state G_s , the raw departure from the safe model is

$$R(f_i, G_s) = P(f_i, G_s) - \hat{P}_{\text{safe}}(f_i, G_s). \quad (3.5)$$

A large common compression component can shift the entire spectrum. To isolate frequency-dependent redistribution, the median residual of each stress state is removed:

$$S(f_i, G_s) = R(f_i, G_s) - \text{median}_f \{R(f, G_s)\}. \quad (3.6)$$

The mean stress-state shape residual is then

$$\bar{S}(f_i) = \frac{1}{N_s} \sum_{G_s \in \mathcal{G}_{\text{stress}}} S(f_i, G_s). \quad (3.7)$$

A robust scale estimate is constructed from the median absolute deviation,

$$\sigma_{\text{MAD}} = 1.4826 \text{ median}_i \left| \overline{S}(f_i) - \text{median}_j \overline{S}(f_j) \right|, \quad (3.8)$$

and the adaptive nonlinear-candidate threshold is

$$T = \max(3 \text{ dB}, 4.5\sigma_{\text{MAD}}). \quad (3.9)$$

For the final V3 dataset, $\sigma_{\text{MAD}} = 5.605 \text{ dB}$ and $T = 25.221 \text{ dB}$.

3.5 Clipping Fraction

The normalized complex IQ samples were scaled so that the nominal ADC endpoints were near ± 1 . A simple clipping indicator was defined as

$$\chi[n] = \begin{cases} 1, & |\Re\{x[n]\}| \geq 0.995 \text{ or } |\Im\{x[n]\}| \geq 0.995, \\ 0, & \text{otherwise,} \end{cases} \quad (3.10)$$

and the clipping fraction of a capture was

$$C = \frac{1}{N} \sum_{n=0}^{N-1} \chi[n]. \quad (3.11)$$

This metric is intentionally direct. It does not attempt to infer the exact internal stage responsible for compression; it indicates when the digitized record spends a significant fraction of time at the numerical rails.

3.6 Band-Power and SNR Estimation

For a target band $\mathcal{B}$ containing K spectral bins, the integrated target power estimate was

$$P_{\text{sig}} = \sum_{f_k \in \mathcal{B}} \widehat{S}_{xx}(f_k) \Delta f. \quad (3.12)$$

A local noise-density estimate N_0 was obtained from a guard-separated set of nearby spectral bins. The equivalent noise power in the target band was

$$P_{\text{noise}} = N_0 K \Delta f, \quad (3.13)$$

and the reported SNR was

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{sig}}}{P_{\text{noise}}} \right). \quad (3.14)$$

Because the ambient FM signal is modulated and the estimator is not a calibrated power meter, these values are used as relative physical-validation metrics rather than absolute receiver sensitivity measurements.

3.7 LO-Coordinate Validation

A fixed receiver artifact may remain at an approximately constant baseband coordinate when the LO changes. A real RF signal instead obeys

$$f_{\text{RF}} = f_{\text{LO}} + f_{\text{BB}}. \quad (3.15)$$

If the same RF signal is measured at multiple LO settings, the detected baseband coordinate should shift by the negative of the LO change while the reconstructed RF coordinate remains constant. The RF tracking standard deviation was therefore

$$\sigma_{\text{RF}} = \sqrt{\frac{1}{M-1} \sum_{m=1}^M \left(\hat{f}_{\text{RF},m} - \bar{f}_{\text{RF}} \right)^2}. \quad (3.16)$$

This test became one of the strongest protections against mistaking a receiver-fixed spur for a genuine over-the-air signal.

Layer 1: Constructive Theory and Simulation

4.1 Purpose of the Constructive Layer

The first layer was designed to answer a limited theoretical question: if a nonlinear receiver produces predictable IM3 artifacts from two hidden parents, can those artifacts carry enough information to reconstruct the parents? It was not designed to replicate the complete analog behavior of an RTL-SDR. The goal was to validate the algebra, the receiver-state dependency, and a local identifiability mechanism under controlled conditions before proceeding to hardware.

4.2 Simulation Parameters

The discrete-time model used $f_s = 1.024$ MHz and $N = 65,536$ samples, corresponding to a DFT grid spacing of

$$\Delta f_{L1} = \frac{1.024 \times 10^6}{65536} = 15.625 \text{ Hz.} \quad (4.1)$$

The parent frequencies were $f_1 = 125$ kHz and $f_2 = 200$ kHz, with amplitudes $A_1 = 0.65$ and $A_2 = 0.45$. Their phases were 0.30 and -0.55 rad. The complex input was

$$x[n] = A_1 e^{j(2\pi f_1 n / f_s + \phi_1)} + A_2 e^{j(2\pi f_2 n / f_s + \phi_2)}. \quad (4.2)$$

The nonlinear coefficients were $c_{10} = 1$, $c_{20} = 0.075$, $c_{02} = 0.055$, and $c_{21} = 0.180$, with a very small forward-model noise term used for the nominal validation run.

The expected third-order difference frequencies were

$$g_1 = 2(125) - 200 = 50 \text{ kHz,} \quad (4.3)$$

$$g_2 = 2(200) - 125 = 275 \text{ kHz.} \quad (4.4)$$

Additional expected components included the parent fundamentals, second harmonics, the sum frequency, and the absolute difference frequency.

4.3 State-Spectroscopy Sweep

To demonstrate state dependence, the third-order coefficient was swept over

$$c_{21} \in \{0.045, 0.075, 0.110, 0.150, 0.180, 0.230, 0.300\}. \quad (4.5)$$

The amplitudes measured at g_1 and g_2 were correlated with c_{21} . Both recorded correlations exceeded the gate value of 0.9999 and were effectively unity. This result established that the nonlinear observables were controllably state dependent in the constructive model.

4.4 Frequency and Amplitude Reconstruction

The observed products occurred exactly on the expected DFT bins at 50 kHz and 275 kHz. Applying eqs. (2.11) and (2.12) recovered 125 kHz and 200 kHz with zero grid-level frequency error. The product amplitudes also contain information about the parent amplitudes. In the adopted complex cubic model, the two product magnitudes were normalized by $|c_{21}|$ and combined through a ratio-product construction. The recovered amplitudes were approximately 0.650000011 and 0.449999998, corresponding to a maximum relative amplitude error of approximately $1.7 \times 10^{-6}\%$ in a numerical re-evaluation of the recorded model.

4.5 Jacobian Rank Gain

A numerical Jacobian was evaluated with respect to the two parent amplitudes while the selected observation consisted only of the IM3-frequency measurements. Under the intentionally linear masked model, the selected observation did not depend on the parent amplitudes at those bins and the Jacobian rank was zero. Under the nonlinear model, both amplitudes affected the IM3 observables and the rank increased to two. Therefore

$$\text{rank } \mathbf{J}_{\text{linear}} = 0, \quad \text{rank } \mathbf{J}_{\text{nonlinear}} = 2. \quad (4.6)$$

The result is significant as a constructive example but should not be generalized beyond the chosen support and observation mask.

4.6 Monte Carlo Robustness

Noise robustness was evaluated over input SNR values from -10 to 30 dB in 5 dB steps, with 40 trials per SNR. A trial was counted as successful only if the maximum parent-frequency error was within 1.5 DFT bins and the weaker local IM3 SNR was at least 6 dB. The recorded

Table 4.1: Layer 1 gate summary.

Gate	Result
Forward frequency validation	PASS
Receiver-state reproducibility	PASS
Hidden-frequency reconstruction	PASS
Hidden-amplitude reconstruction	PASS
Identifiability rank gain	PASS
Monte Carlo robustness	PASS
Blind parent recovery	PASS

campaign achieved 100% recovery at -5 dB and above; the lowest SNR point showed reduced recovery. The Layer 1 gate required at least 95% recovery for SNR values of 10 dB and above and passed.

4.7 Interpretation of Layer 1

Layer 1 establishes three things and deliberately does not establish a fourth. It establishes that the algebraic parent-frequency map is invertible, that a state-dependent nonlinear term can create measurable information-bearing artifacts, and that the selected nonlinear observation can recover local rank that is absent from the selected linear masked observation. It does not prove that an arbitrary physical receiver provides the same operator, that ambient spurs are causal IM3 products, or that the resulting inverse problem is globally unique. Those claims require physical measurement, which motivated the subsequent campaigns.

Experimental Platform and Data-Acquisition Methodology

5.1 Hardware Platform

The physical experiments used an RTL-SDR Blog V4 receiver. The manufacturer's V4 datasheet identifies the R828D tuner and RTL2832U 8-bit ADC, gives a 500 kHz to 1.766 GHz tuning range, and lists 2.56 MHz as a stable bandwidth with higher rates possible at the risk of dropped samples [21]. The selected sample rate of 2.048 MS/s therefore remained within the stated stable-bandwidth region. The same datasheet describes a 50-ohm nominal input, a 1-ppm TCXO, and a triplexed HF/VHF/UHF front end [21]. The official V4 documentation also emphasizes that updated V4-compatible drivers are required [22, 23].

Before the final campaign, command-line characterization identified the connected unit as an RTL-SDR Blog V4 with a Rafael Micro R828D tuner. The manual tuner-gain list exposed by the driver included values from 0 to 49.6 dB. The measurements in this thesis used nonzero manual gain values so that the tuner state was commanded explicitly rather than left in automatic gain mode.

5.2 Receive Environment

The final physical measurements were passive, receive-only observations of the commercial FM broadcast band. The receiver was connected to an external antenna and positioned to obtain strong real-world broadcast signals. No RF transmission was performed. This distinction matters because the final IM3 search relied on naturally occurring ambient carriers rather than controlled laboratory tones. The environment therefore provided realism at the cost of causal control.

5.3 Acquisition Evolution

The acquisition subsystem required substantial validation because a nonlinear experiment is only meaningful if changes in the stored IQ data correspond to changes in the receiver state.

Early attempts using a high-level SDR interface produced suspiciously flat state responses. Those datasets were rejected rather than interpreted. Subsequent command-line experiments showed that direct RTL-SDR capture could produce full-length changing IQ records and physically plausible gain dependence.

Several intermediate acquisition paths were also rejected. Synchronous command-line capture could block on some Windows runs, and an attempted persistent TCP architecture stalled before reliable streaming. These failures did not contribute data to the final scientific result. The final working path returned to the simplest previously validated command-line acquisition: direct asynchronous capture through the V4-compatible RTL-SDR driver. A capture was accepted only when the expected binary file size was produced and the data passed subsequent integrity checks. The final campaign used 250,000 complex samples per capture, corresponding to 500,000 unsigned 8-bit I/Q bytes and a nominal record duration of 0.122070 s.

5.4 Final Survey Geometry

The RF survey covered 88–108 MHz with overlapping centers. The nominal center spacing was 0.700 MHz. Twenty-eight center frequencies were used, beginning at 88.720 MHz and ending at 107.280 MHz; the final edge center was adjusted so that the union of usable subbands closed the upper boundary. The adopted usable half-band was 780 kHz with a 55 kHz DC exclusion around each center. The geometric coverage was evaluated before the final run as

$$\eta_{\text{cov}} = \frac{\mu(\bigcup_c \mathcal{B}_c)}{108 - 88}, \quad (5.1)$$

where μ is interval measure in MHz and $\mathcal{B}_c$ is the accepted RF region of center c after the DC guard is removed. The final geometry predicted and achieved 100% coverage on the analysis grid.

5.5 Receiver States

The final survey used five gain states,

$$\mathcal{G} = \{8.7, 14.4, 19.7, 29.7, 33.8\} \text{ dB}. \quad (5.2)$$

The safe set was

$$\mathcal{G}_{\text{safe}} = \{8.7, 14.4, 19.7\} \text{ dB}, \quad (5.3)$$

and the stress set was

$$\mathcal{G}_{\text{stress}} = \{29.7, 33.8\} \text{ dB}. \quad (5.4)$$

Gain order was randomized within each survey center to reduce systematic coupling between receiver state and slow environmental variation.

5.6 Capture Integrity and Non-Stale Validation

A recurring experimental risk was stale or repeated data. The final methodology therefore treated file length as necessary but not sufficient. Selected byte subsets from successive IQ files were compared, and the campaign was required to show a substantial difference between fingerprints. The final V3 survey passed this non-stale test. This gate is methodologically important because an apparently perfect state-response curve produced from repeated data would be meaningless.

5.7 Spectral Estimation

Each IQ record was mean removed before spectral analysis to suppress the strongest DC term. Windowed FFT segments were averaged to obtain a stable spectral estimate. The resulting center-relative frequency grid was transformed into RF coordinates by

$$f_{\text{RF},k} = f_c + f_{\text{BB},k}. \quad (5.5)$$

Contributions from overlapping survey centers were accumulated in linear power and averaged on a common 10-kHz RF grid. The stitched representation was therefore an overlap-averaged multi-state spectrum rather than a concatenation of arbitrary screenshots.

5.8 Gate-Based Experimental Logic

The experimental program was organized as a sequence of gates. This was adopted to prevent later nonlinear analyses from being performed on invalid earlier stages. The final hierarchy was:

1. acquisition smoke capture;
2. stitched survey coverage;
3. non-stale IQ;
4. observable real RF target;
5. nonlinear spectral redistribution;
6. receiver compression;

7. real-RF LO tracking;
8. ambient intermodulation graph;
9. product-only parent reconstruction.

The first seven gates ultimately passed. The last two failed, leading to partial physical validation rather than a full inversion claim.

Experimental Campaign Evolution

6.1 Why Multiple Campaigns Were Necessary

The physical validation did not proceed directly from theory to a successful final dataset. Each campaign exposed a different failure mode and led to a stricter subsequent design. Reporting this progression is important because the final metrics are meaningful partly because the earlier weaknesses were explicitly identified and corrected.

6.2 Campaign A: Initial Real-World Engine

The first strong real-world campaign used 15 survey centers, two survey gain states of 8.7 and 42.1 dB, 12 targeted gain states, a 2.048 MS/s sample rate, and 400,000 complex samples per capture. The selected target was 95.100000 MHz with 30.924 dB prominence. The target varied strongly with gain: the measured target-power span was 34.4737 dB and the gain-target correlation was 0.949944. Repeatability was already good, with a median two-repeat power difference of 0.1524 dB. LO-coordinate testing was particularly successful: the target RF tracking standard deviation was 0.044 kHz and the maximum baseband tracking error was 0.431 kHz.

However, two decisive weaknesses remained. The two-state survey covered only 83.8581% of the requested band, and high-gain operation produced severe clipping. The final high-gain states exhibited clipping fractions of roughly 56–60% in the strongest region. The campaign therefore failed the coverage and ADC-dynamic-range gates. It also produced no accepted nonlinear candidate bins and no intermodulation graph. The correct conclusion at this stage was not that the concept had failed, but that the measurement design could not separate a safe response from a stress response with sufficient coverage and dynamic-range control.

6.3 Campaign B: Rigorous V2

The second campaign introduced a three-state wideband survey and separated a safe response from a stress response. Twenty-four centers at 0.8 MHz spacing provided 100% coverage

in the adopted stitched geometry. The survey states were 8.7, 25.4, and 29.7 dB, while the targeted gain sweep again used 12 states.

This campaign selected a 90.790000 MHz target with 29.559 dB prominence. The safe-region gain-target correlation was 0.994199907 and the safe linear coefficient of determination was $R^2 = 0.988433455$. No clipping was observed in the safe region, whereas the maximum stress clipping fraction reached 0.658736067. The response departed from the safe model at a 33.8 dB gain command in the targeted sweep and accumulated a compression-depth metric of 15.683 dB. The best target SNR was 31.948 dB at a 29.7 dB gain command. The median repeat difference was 0.093 dB.

The frequency-dependent residual analysis produced a spectral-shape RMS of 8.857 dB and three wideband nonlinear candidates. A separate targeted search produced no accepted nonlinear candidates, and the intermodulation graph remained unestablished. LO validation remained strong: RF tracking standard deviation was 0.462756 kHz, maximum baseband tracking error was 9.959322 kHz, and median LO-validation SNR was 8.095 dB. All twelve hardware/measurement gates in this campaign passed. For the first time, receiver nonlinear compression and nonlinear spectral redistribution were both labeled established, while IM3 graph and hidden-parent reconstruction remained not established.

6.4 Campaign C: Final V3

The final campaign was redesigned around five survey gain states and a stricter safe/stress partition. The center spacing was reduced to 0.700 MHz after an intermediate 0.800 MHz geometry produced only 95.552% coverage. This correction is a useful example of why geometric coverage should be computed before acquisition rather than inferred after the fact.

The final acquisition also adopted a simple asynchronous command-line path that had previously demonstrated reliable complete IQ files. A driver cleanup return code that occasionally accompanied a complete 500,000-byte asynchronous capture was normalized only when the full expected file existed and the log showed no fatal transfer error. This prevented a complete physical capture from being discarded because of a post-capture USB cleanup code while still rejecting wrong-length or fatally logged captures.

6.5 Methodological Lessons from the Campaign Sequence

Three lessons were carried into the final interpretation. First, strong gain dependence is not sufficient evidence of useful nonlinearity; it can result from ordinary gain scaling or gross clipping. Second, a wideband nonlinear candidate is not automatically an IM3 product. A separate physical validation stage is required. Third, RF-coordinate tracking is a powerful

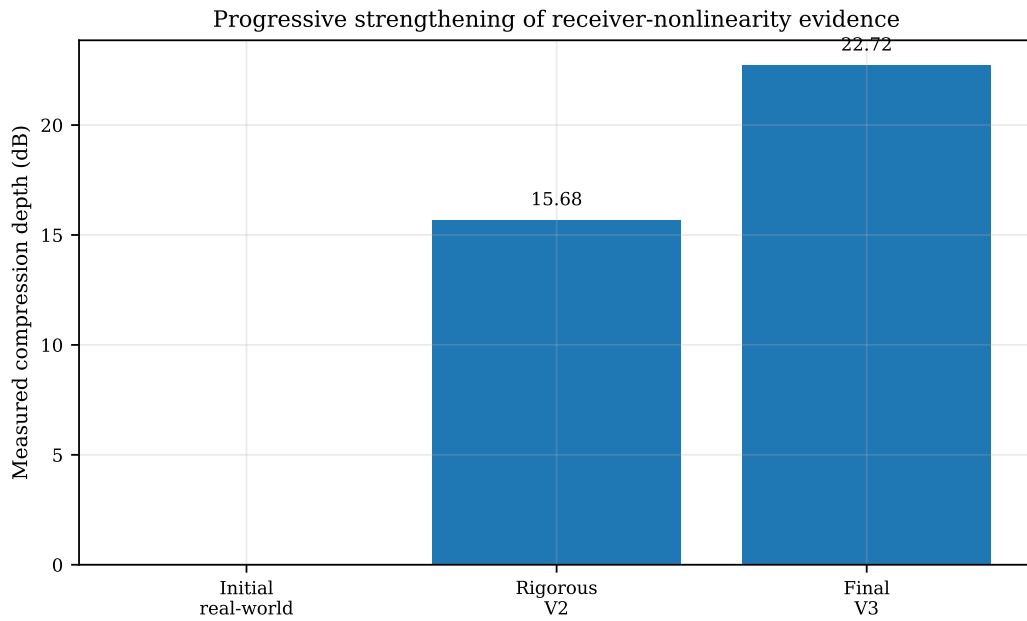


Figure 6.1: Progressive strengthening of the measured receiver-compression evidence across the physical campaigns. Campaign A established strong gain dependence but did not yet provide a rigorous safe/stress compression-depth metric.

Table 6.1: Progression of key campaign-level metrics.

Metric	Campaign A	Campaign B	Final V3
Survey coverage	83.8581%	100%	100%
Selected target (MHz)	95.100000	90.790000	95.097007
Target prominence (dB)	30.924	29.559	34.339
Gain-target correlation	0.949944	0.994200	0.987076
Safe-region R^2	–	0.988433	0.974319
Compression onset (gain dB)	qualitative	33.8	25.4
Compression depth (dB)	–	15.683	22.719
RF tracking std (kHz)	0.044	0.463	0.244
Accepted wideband nonlinear bins	0	3	2
Ambient IM graph	FAIL	FAIL	FAIL

falsification tool. A genuine external RF feature must move through baseband exactly as the LO is shifted, while a receiver-fixed artifact generally does not obey the same transformation.

Final V3 Wideband Survey Results

7.1 Survey Completion and Coverage

The final V3 run completed all 28 survey centers and all five gain states. Every accepted survey file contained the expected 500,000 IQ bytes. Coverage was reported as

$$\eta_{\text{cov}}(G) = 100\%, \quad \forall G \in \mathcal{G}. \quad (7.1)$$

The non-stale-IQ gate also passed, demonstrating that the files were not repeated snapshots. This pair of results closes two of the most important acquisition-validity questions: the requested RF band was fully represented on the stitched grid, and the underlying byte streams changed between captures.

7.2 Clipping Topology Across the FM Band

The clipping distribution was strongly frequency dependent. At the three safe states, clipping was essentially zero over nearly the entire band. At 29.7 and 33.8 dB, clipping appeared primarily around strong FM stations and could exceed 50%. For example, at the 95.020 MHz survey center, the clipping fraction rose from zero at 19.7 dB to 0.4516 at 29.7 dB and 0.6322 at 33.8 dB. At 90.820 MHz, the 33.8 dB state reached 0.6018 while the safe states remained at zero.

This topology is important because it demonstrates that the receiver does not enter stress uniformly across the band. The same gain command can be safe in a weak local spectrum and severely nonlinear near a strong station. Receiver state must therefore be interpreted jointly with the incident spectral environment.

7.3 Target Selection

The strongest suitable final target was identified at

$$f_{\text{target}} = 95.097007 \text{ MHz} \quad (7.2)$$

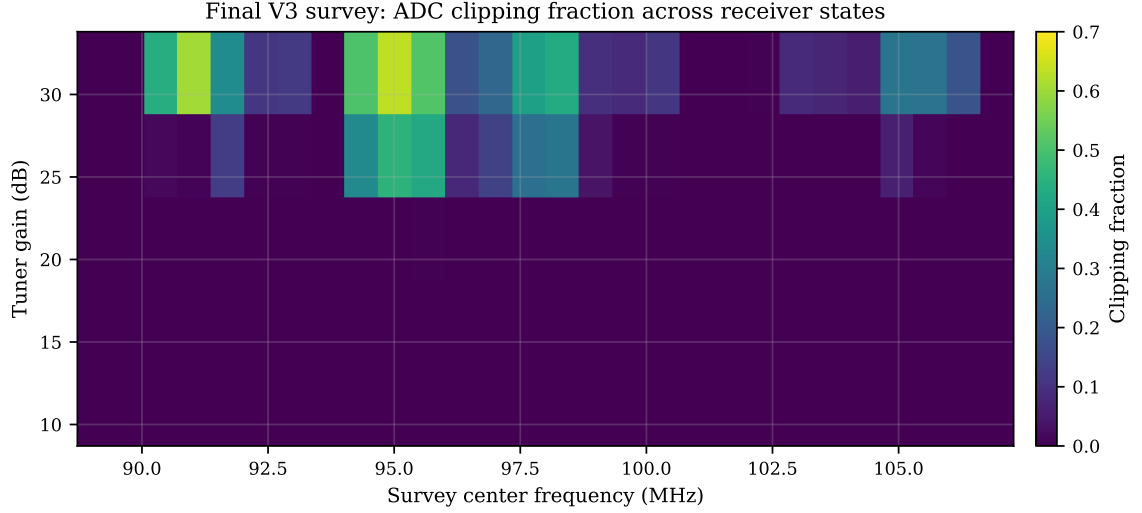


Figure 7.1: Final V3 clipping fraction across the 28 survey centers and five receiver gain states. The safe states are largely unclipped, while high-gain stress states clip strongly near powerful broadcast signals.

with a prominence of 34.339 dB relative to the low-gain spectral floor. The target was subsequently used for the detailed gain sweep and LO-coordinate validation.

7.4 Frequency-Dependent Spectral Redistribution

The safe-state linear model was fit independently at each common RF grid point. Stress-state residuals were then median centered according to eq. (3.6). The robust residual scale was

$$\sigma_{\text{MAD}} = 5.605 \text{ dB}, \quad (7.3)$$

and the threshold from eq. (3.9) was

$$T = 25.221 \text{ dB}. \quad (7.4)$$

Two RF bins exceeded this threshold. Under the adopted gate, nonlinear spectral redistribution was therefore established.

This result should be interpreted carefully. It demonstrates frequency-dependent departure from the safe-state spectral model after common stress response is removed. It does not by itself assign a physical mechanism to each bin. Possible contributors include nonlinear mixing, clipping-generated regrowth, state-dependent filtering, quantization interaction, or combinations of these effects. The later IM3 stage was designed specifically to test whether some of the candidate structure could be linked to two-parent third-order relations.

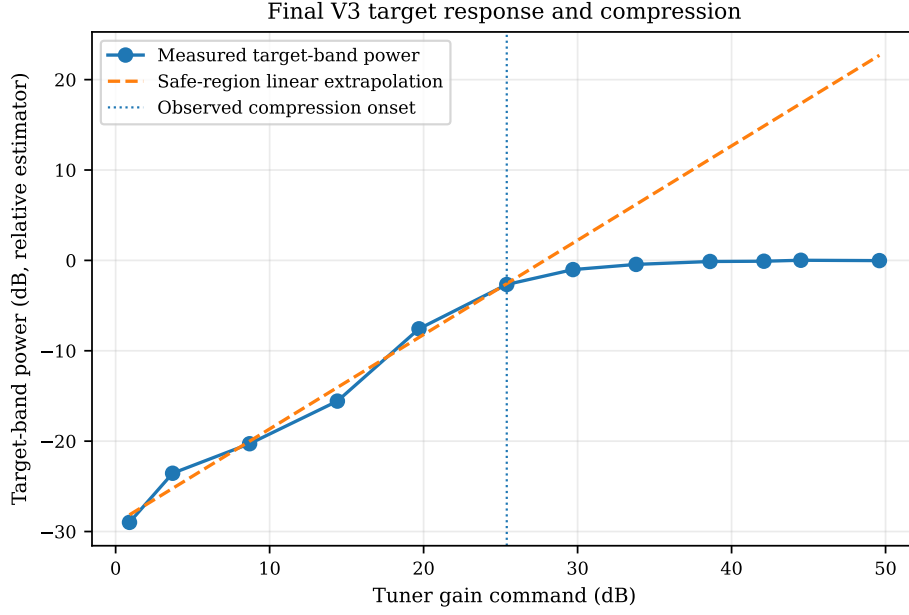


Figure 7.2: Final V3 target-band response versus tuner gain. The dashed line is the low-stress linear extrapolation. The flattening above the observed 25.4 dB onset demonstrates strong receiver-state compression. This is a tuner-gain-state metric, not a calibrated P1dB measurement.

7.5 Target Gain Sweep

The final target gain sweep used twelve manual gain commands and two independent captures per state. The measured target-band power increased rapidly through the safe region, then flattened as clipping rose. The safe-region fit produced

$$R_{\text{safe}}^2 = 0.974319, \quad \rho_{G,P} = 0.987076. \quad (7.5)$$

The first observed departure satisfying the onset criterion occurred at a tuner-gain command of 25.4 dB. Relative to the extrapolated safe-region response, the deepest stress-state departure was

$$D_{\text{comp}} = 22.719 \text{ dB}. \quad (7.6)$$

The maximum measured clipping fraction across the stress sweep was 0.686344. The median absolute difference between the two repeated power measurements at each gain state was only 0.074 dB.

7.6 SNR-Clipping Trade-off

The target SNR increased with tuner gain through the safe region, reaching approximately 33 dB around the 19.7 dB state. At 25.4 dB, clipping exceeded roughly 10% while SNR fell to approximately 26.6 dB. At 29.7 dB, clipping reached about 35% and SNR fell to roughly

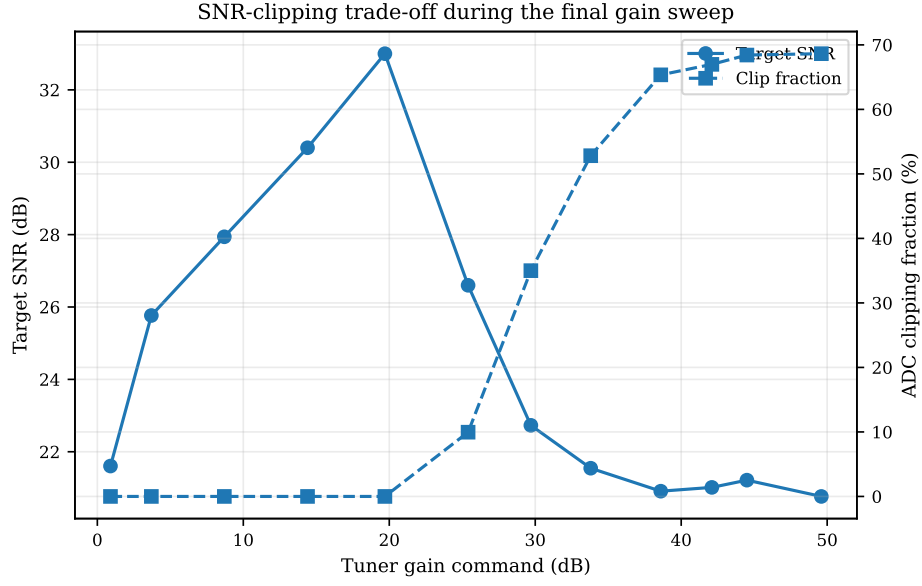


Figure 7.3: Final V3 SNR and clipping fraction versus tuner gain. The best operating region for the selected target is below the strong-clipping stress states.

22.7 dB. Above 33.8 dB the digitized record was clipped for more than half of the samples and the target SNR remained around 20–21 dB.

The SNR decline at high gain reinforces the distinction between “more gain” and “more information.” Once the receiver is driven into heavy clipping, target-band power may numerically increase or saturate while useful signal quality degrades.

7.7 LO-Coordinate Validation

The target was then measured at three LO settings with two repeats each. The detected RF frequencies remained near 95.101 MHz while the baseband coordinate changed as expected. The final summary was

$$\sigma_{\text{RF}} = 0.244 \text{ kHz}, \quad (7.7)$$

$$e_{\text{BB,max}} = 3.997 \text{ kHz}, \quad (7.8)$$

$$\widetilde{\text{SNR}}_{\text{LO}} = 7.523 \text{ dB}. \quad (7.9)$$

The real-RF LO-tracking gate passed.

7.8 Final Receiver-Physics Assessment

By the end of the final V3 receiver-state stage, seven physical and measurement gates had passed: acquisition, coverage, non-stale IQ, real RF target, nonlinear spectral redistribution,

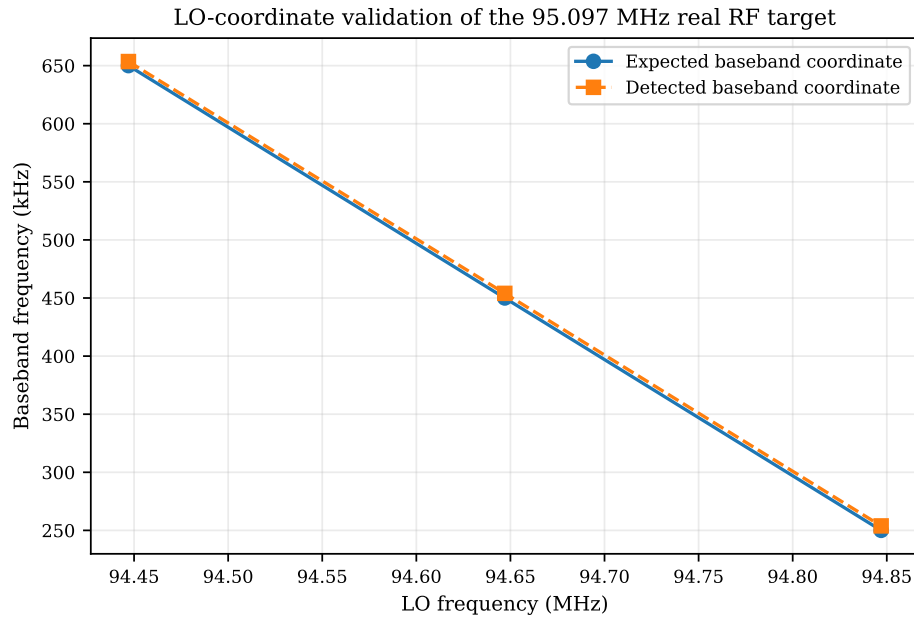


Figure 7.4: Expected and measured baseband coordinate of the selected target as the LO is shifted. The reconstructed RF coordinate remains stable, supporting interpretation as a real external RF signal.

receiver compression, and LO tracking. This is sufficient to state that receiver-state nonlinearity was physically observed and repeatably characterized under strong ambient FM excitation. It is not sufficient to state that specific ambient spectral features are causal IM3 products. That stronger question is treated separately in the next chapter.

Ambient IM3 Graph and Product-Only Reconstruction Attempt

8.1 Candidate-Graph Construction

Parent candidates were extracted from the stitched survey. For each parent pair (f_1, f_2) , the predicted IM3 difference frequencies were calculated using eqs. (2.7) and (2.8). Candidate products outside the 88–108 MHz survey region were rejected. Predicted products too close to known parent peaks were also excluded to reduce ambiguity between a nonlinear product and an ordinary broadcast station.

The final screening generated 18 geometrically valid candidate pairs. The four highest-priority candidates were subjected to fresh validation. The purpose of the fresh stage was to avoid circular reasoning: the same survey bins used to propose a pair were not accepted as proof that the pair was physically real.

8.2 Physical Product Gates

A candidate product was required to behave like a real RF feature. The fresh validation therefore evaluated product SNR, reconstructed RF stability under LO shifts, and maximum baseband-coordinate error. In the robust continuation stage, a product was not accepted merely because its observed frequency was numerically near the prediction.

For interpretation, a valid product would ideally satisfy

$$\text{SNR}_{g_i} \geq 3 \text{ dB}, \quad (8.1)$$

$$\sigma_{\text{RF},g_i} \leq 30 \text{ kHz}, \quad (8.2)$$

and

$$e_{\text{BB},g_i} \leq 50 \text{ kHz}. \quad (8.3)$$

The exact numerical thresholds are engineering gates chosen for this campaign, not universal standards.

8.3 Pair 1

The first pair used parent candidates at 95.097007 and 98.494559 MHz, predicting products at 91.699455 and 101.892111 MHz. The observed frequencies were 91.648581 and 101.898316 MHz. Algebraic inversion produced reconstructed parents at 95.065159 and 98.481738 MHz, giving a maximum parent error of only 31.848 kHz.

The physical evidence, however, was poor. The first product had RF-coordinate standard deviation 60.255 kHz, maximum baseband error 65.230 kHz, and SNR -20.435 dB. The second product tracked much more stably in coordinate, with 0.279 kHz RF standard deviation and 6.399 kHz baseband error, but its SNR was still only -19.491 dB. Because both products did not satisfy the physical gates, the intermodulation graph failed despite the small algebraic reconstruction error.

8.4 Pairs 2–4

Pair 2 used parents at 95.489806 and 96.394546 MHz and predicted products at 94.585067 and 97.299285 MHz. The observed products were 94.520698 and 97.232392 MHz. The product RF standard deviations were 4.076 and 4.656 kHz, but the baseband errors were 71.418 and 70.383 kHz. The SNR values were -5.474 and -20.407 dB. Reconstruction error was 66.051 kHz. The pair failed.

Pair 3 used parents at 96.394546 and 98.197155 MHz, predicting 94.591937 and 99.999764 MHz. The observed frequencies were 94.523461 and 99.998798 MHz. RF-coordinate standard deviations were 2.040 and 2.644 kHz. The first product had a 71.978 kHz baseband error and -6.465 dB SNR; the second product had a much smaller 5.257 kHz baseband error but an even lower -23.961 dB SNR. Reconstruction error was 45.972 kHz. The pair failed.

Pair 4 used parents at 95.097007 and 98.197155 MHz and predicted products at 91.996859 and 101.297302 MHz. The observed products were 92.075781 and 101.369122 MHz. RF-coordinate stability was numerically good, with 0.933 and 1.245 kHz standard deviations, but the baseband errors were 79.834 and 72.491 kHz and the SNR values were -12.370 and -21.643 dB. Reconstruction error was 76.554 kHz. The pair failed.

8.5 Why Algebraic Consistency Was Rejected as Proof

The strongest methodological result of the IM3 stage is the rejection of false confidence. The inversion matrix in eq. (2.10) is well conditioned and linearly maps two measured frequencies into two reconstructed parent frequencies. In a dense FM spectrum, many unrelated peaks

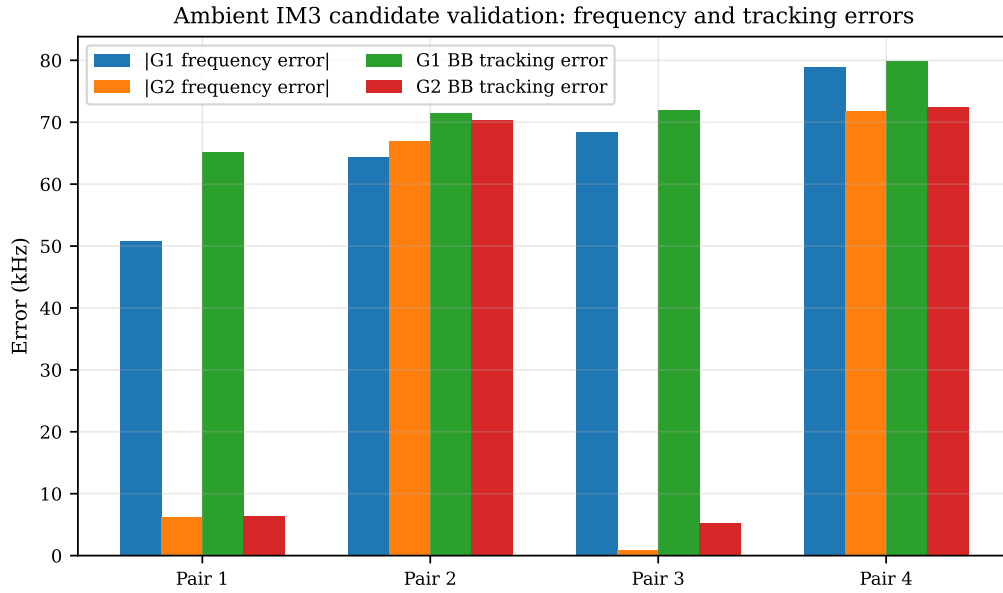


Figure 8.1: Frequency and LO-coordinate errors of the four final ambient IM3 candidate pairs. Several candidate peaks were close to the algebraic prediction but did not behave well enough under the physical tracking gates.

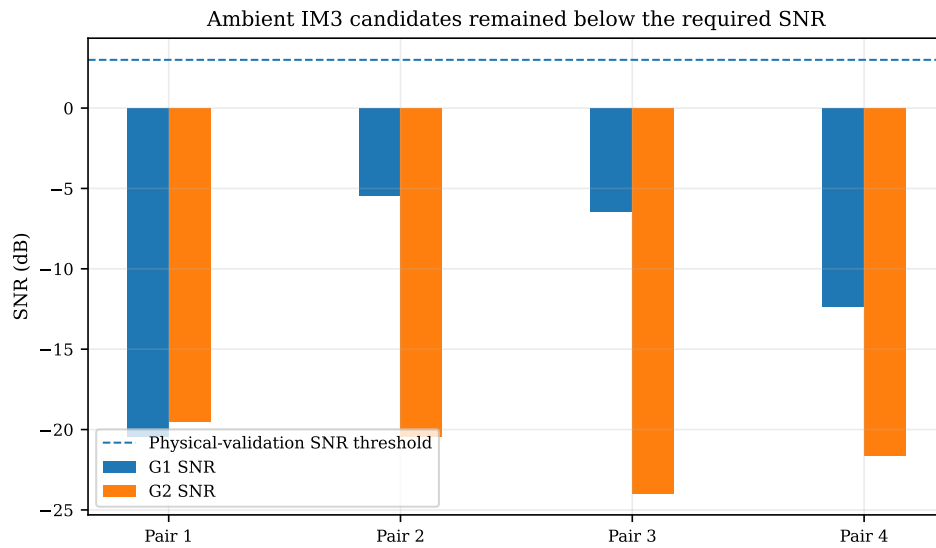


Figure 8.2: SNR of the fresh candidate products. Every tested product remained below the adopted positive-SNR physical validation threshold; many were deeply below the estimated local noise level.

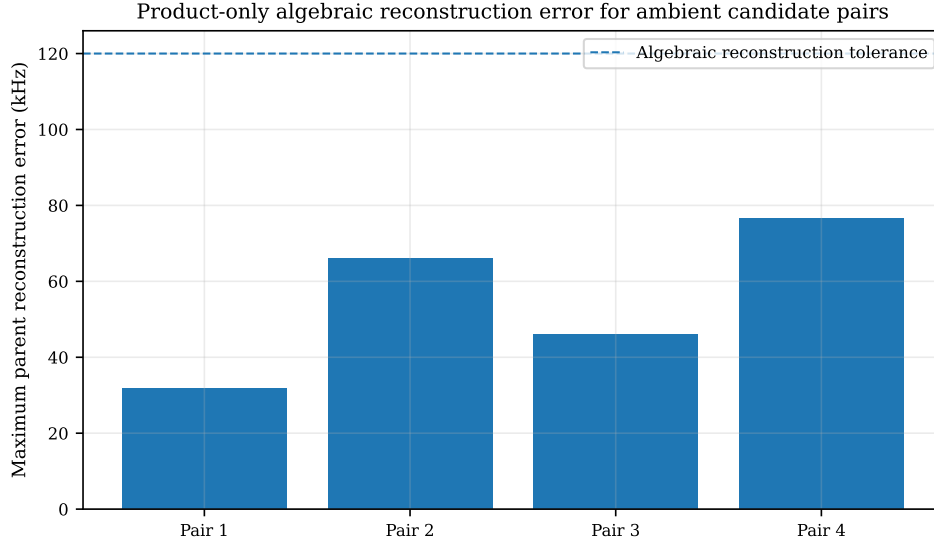


Figure 8.3: Algebraic product-only parent reconstruction errors for the four final candidate pairs. These errors are numerically modest, but the corresponding product signals failed physical SNR/tracking gates and therefore cannot be accepted as causal IM3 evidence.

can accidentally lie near algebraically compatible coordinates. Therefore, an error such as 31.848 kHz is not convincing if the candidate product itself has SNR near -20 dB.

The experiment consequently treats physical observability as a prerequisite for inversion. This is analogous to parameter estimation more generally: an estimator may return a numerical value even when the data do not contain sufficient information to support that estimate [14, 19]. The final thesis therefore reports the small algebraic errors but does not convert them into a hidden-parent claim.

8.6 Final IM3 Assessment

All four fresh candidate pairs failed the intermodulation graph. Because product-only reconstruction was conditioned on a valid graph, all four also failed the parent-reconstruction gate. The final result is

$$\boxed{\text{Ambient IM3 graph: NOT ESTABLISHED}} \quad (8.4)$$

and

$$\boxed{\text{Product-only parent reconstruction: NOT ESTABLISHED}}. \quad (8.5)$$

This negative result is retained as part of the core contribution rather than hidden as an inconvenient outcome.

Uncertainty, Robustness, and Threats to Validity

9.1 Frequency Resolution and Centroid Uncertainty

The 8192-point FFT at 2.048 MS/s gives 250-Hz bins, but the true measurement uncertainty is not 250 Hz. Broadcast FM is modulated, the spectral centroid is estimated over a finite window, multipath and fading alter the local spectrum, and receiver tuning may have finite frequency error. The LO test provides a more empirical estimate of coordinate repeatability for the selected target: 0.244 kHz RF standard deviation across the three LO states. Product candidates with tens of kilohertz of tracking error were therefore clearly less stable than the target.

9.2 Amplitude and Power Calibration

The target-power metric is derived from normalized unsigned 8-bit IQ samples and is not calibrated to absolute dBm at the antenna connector. The resulting values are suitable for within-experiment state comparison but not for direct comparison with laboratory power-meter readings. For this reason, the thesis reports “target-band power” and “compression depth relative to safe-state extrapolation” rather than absolute gain compression at a known input level.

9.3 Eight-Bit Quantization and Clipping

The RTL2832U ADC is 8-bit according to the V4 datasheet [21]. Quantization is therefore non-negligible compared with modern high-resolution SDRs. More importantly, the final stress states intentionally entered regions where the normalized samples frequently reached the numerical rails. At the strongest states, clipping fractions above 0.68 were observed. Such records are strongly nonlinear and contain severe waveform deformation. Their usefulness is not assumed; they are used to characterize where the receiver leaves its safe operating region.

9.4 Ambient-Scene Nonstationarity

Commercial FM carriers are not controlled two-tone sources. Their instantaneous modulation, transmitter power stability, propagation, local multipath, and interference can vary between sequential captures. Randomizing gain order reduced but did not eliminate this problem. Fresh product validation also measured candidate frequencies sequentially rather than simultaneously. This is one of the main reasons the thesis refuses to interpret a weak candidate peak as causal IM3 without strong SNR and LO-coordinate evidence.

9.5 Receiver Internal Architecture

The receiver was treated as a black-box measurement system. The experiments do not isolate whether a particular nonlinear feature originated in the R828D tuner, an amplifier stage, mixer, ADC, power-supply interaction, or another internal mechanism. The manufacturer’s documentation describes the V4 front-end filtering and tuner architecture [21, 22], but the present experiment does not probe internal nodes. Consequently, the claim is at the receiver-system level.

9.6 Safe/Stress Model Dependence

The fitted safe-region line depends on the chosen gain states. The final safe set used 8.7, 14.4, and 19.7 dB because these states remained essentially unclipped over the selected target. A different target or RF environment could require a different safe set. The threshold $T = \max(3, 4.5\sigma_{\text{MAD}})$ is likewise an engineering detection rule, not a universal statistical significance level. The robustness comes from its conservative magnitude and from the fact that the final physical claim does not depend on this threshold alone; compression, clipping, repeatability, and LO tracking provide independent evidence.

9.7 Multiple-Hypothesis Risk in IM3 Search

Testing many possible parent pairs raises a multiple-hypothesis problem. Even if each predicted product frequency has a low probability of accidental proximity to a peak, the probability of at least one apparent match increases with the number of pairs. The final workflow partially mitigated this risk by fresh validation and strict physical gates, but it did not compute a formal family-wise false-alarm probability. This is another reason the thesis does not accept algebraic proximity as proof.

9.8 Reproducibility

The final campaign is reproducible at the methodological level because the sample rate, capture length, RF band, center spacing, gain states, safe/stress partition, clipping rule, spectral grid, target metrics, robust residual rule, and IM3 gates are documented. Exact replication of the measured numbers is not expected in a different location or time because the experiment used ambient broadcasters. A controlled laboratory replication would be required for location-independent causal verification.

Discussion

10.1 What Was Established

The most defensible conclusion is that the RTL-SDR Blog V4 receiver exhibited strong, repeatable, receiver-state-dependent nonlinear behavior under the measured ambient FM conditions. The evidence is not a single plot. It is the joint result of a validated acquisition chain, full survey coverage, non-stale IQ, a real RF target verified through LO-coordinate motion, a highly linear low-stress response, a pronounced high-stress departure, rapidly increasing clipping, SNR degradation after the optimum region, and frequency-dependent residual bins after common compression was removed.

The target's safe-region R^2 of 0.974319 and correlation of 0.987076 show that the low- and moderate-gain points form a coherent trend. The measured curve then ceases to follow the extrapolation: by the highest states, the linear model predicts continued growth while the digitized target power remains near saturation. The 22.719-dB compression-depth metric is therefore large relative to the 0.074-dB repeat difference. The phenomenon cannot plausibly be explained by repeatability noise alone.

10.2 What Was Not Established

The thesis does not establish that the two residual bins detected in the wideband stress analysis are IM3 products. It also does not establish that the four tested ambient pair graphs correspond to causal mixing of the proposed parents. Candidate products were too weak and, in several cases, failed the LO-coordinate error threshold. The final hidden-parent reconstruction claim is therefore not made.

This restraint strengthens rather than weakens the receiver-nonlinearity result. A project seeking only a positive headline could have stopped after the algebraic reconstruction errors of 31–77 kHz. The present methodology instead asks whether the “products” are physically observable enough to deserve the name. They were not.

10.3 Interpretation of Nonlinear Spectral Redistribution

The two final nonlinear spectral bins should be regarded as evidence of frequency-selective departure from the safe-state transfer surface. At high stress, clipping alone generates wide spectral consequences because a time-domain limiter redistributes energy across frequency. Mixer and amplifier nonlinearities can create structured intermodulation, while filters and gain stages can change the relative prominence of components. The observed residual bins are therefore consistent with a nonlinear receiver but not uniquely diagnostic of a cubic mechanism.

This distinction aligns with the caution in nonlinear RF literature that low-order power-series models are useful for weak nonlinearity but become insufficient in strongly clipped operation [2]. In the final campaign, clipping fractions above 50% at several high-gain states clearly place those records beyond a purely weak cubic regime.

10.4 Receiver-State Spectroscopy as a General Method

Even without ambient parent reconstruction, receiver-state spectroscopy remains useful as a characterization method. A conventional single-state spectrum does not reveal how a component changes when the receiver is moved through its gain range. The surface $\mathcal{V}(f, G)$ exposes where signals remain linear, where the ADC begins to clip, where SNR reaches an optimum, and where particular spectral regions respond differently from the common-mode gain trend.

This suggests a broader receiver-design principle: receiver settings should not be viewed only as nuisance parameters. When their effect is calibrated, they can become experimental variables. The present thesis demonstrates this principle for gain state. Other potential state dimensions include LO placement, analog filter selection, RF-path selection, bias conditions in reconfigurable hardware, or intentionally inserted attenuation.

10.5 Relation to Standard RF Metrics

The observed behavior is related to but distinct from standard compression and IP3 metrics. In a laboratory amplifier test, the input power is calibrated and swept while the device gain and output power are measured. Here the external ambient scene was not power swept. Instead, tuner gain was changed internally. The resulting “compression onset at 25.4 dB” therefore means that, for the selected ambient target and receiver configuration, the measured target response began to depart from the low-stress gain-state extrapolation at that command. It does not mean the receiver has an input P1dB of 25.4 dBm or any other input power.

Similarly, standard IM3/IP3 theory predicts a slope-three third-order response under controlled two-tone excitation [1, 2]. No such slope was measured here because the parent input powers were not controlled. The ambient IM3 stage should therefore be understood as a graph-consistency and physical-tracking test, not an IP3 test.

10.6 Scientific Value of the Negative IM3 Result

The negative result places a clear boundary around what the present hardware setup can support. Under passive ambient FM reception, receiver nonlinearity is easy to drive and measure, but reliable attribution of weak nonlinear products to a specific parent pair is much harder. Dense broadcast spectra create coincidence risk, and sequential captures make weak product validation vulnerable to environmental variation. The data show that the algebraic inversion can look attractive even when product SNR is poor. This is precisely the regime in which strict gate logic is most valuable.

Future Experimental Closure

11.1 Controlled Two-Tone Validation

The natural next experiment is a conducted, receive-only two-tone test in which two clean RF generators are combined through a 50-ohm network and heavily attenuated before entering the SDR. The target relation would be

$$f_1, f_2 \rightarrow \{2f_1 - f_2, 2f_2 - f_1\}. \quad (11.1)$$

The advantage is causal control: either parent can be switched off, parent spacing can be changed, and the expected product motion is exactly known.

A particularly strong test would shift f_1 by Δf while holding f_2 fixed. The products should then move by

$$\Delta g_1 = 2\Delta f, \quad (11.2)$$

$$\Delta g_2 = -\Delta f. \quad (11.3)$$

Such differential motion is difficult for unrelated spurs to imitate. A calibrated input-power sweep could additionally test the expected slope-three growth in the weakly nonlinear region before compression [1].

11.2 Source-Side Versus Receiver-Side Distortion

A conducted experiment should also discriminate source/combiner distortion from receiver-generated distortion. Inserting an additional attenuator immediately ahead of the receiver reduces the parent levels at the receiver. If the IM3 is generated inside the receiver in the weak cubic regime, reducing both parents by 10 dB should reduce the receiver-generated IM3 products by approximately 30 dB while the fundamentals fall by 10 dB. In contrast, an IM3 product already generated upstream would pass through the same attenuator with approximately the same 10-dB reduction as the parents. This relative-scaling test would provide strong causal localization.

11.3 Blind Product-Only Recovery

After a controlled IM3 graph is verified, the direct parent bins should be masked from the inversion input. The algorithm would then receive only the product coordinates and reconstruct the parents using eqs. (2.11) and (2.12). A fair linear-only baseline would receive the same masked observation without the nonlinear products. This would convert the constructive Layer 1 idea into a controlled physical hidden-observation experiment.

11.4 Statistical Null Testing

Future work should also add an explicit multiple-hypothesis null model. One approach is to circularly shift the stress-residual spectrum relative to the parent-frequency set and recompute the best IM3 pair score thousands of times. If the observed score exceeds, for example, the 99.9th percentile of the null distribution, the probability of an accidental algebraic alignment would be quantitatively constrained. Such a test would complement, not replace, the physical LO and SNR gates.

Conclusion

This thesis investigated whether nonlinear artifacts of a low-cost software-defined receiver could be treated as candidate auxiliary measurement features rather than only as distortion. A constructive nonlinear model first demonstrated the algebraic and local-identifiability mechanism. The physical program then moved through successive real-world campaigns that progressively corrected coverage, clipping, state-separation, acquisition, and artifact-validation weaknesses.

The final RTL-SDR Blog V4 campaign achieved complete 88–108 MHz coverage at five receiver gain states and validated a real 95.097007 MHz target. The low-stress target response was repeatable and approximately linear, with $R^2 = 0.974319$ and gain-response correlation 0.987076. At high receiver gain, the response compressed severely relative to the safe extrapolation; the compression-depth metric reached 22.719 dB and the maximum clipping fraction reached 0.686344. A robust stress-residual analysis identified two frequency-dependent bins above a 25.221-dB threshold. LO-coordinate validation held the selected RF target stable to 0.244 kHz standard deviation. These results establish receiver-state nonlinear compression and spectral redistribution under the stated measurement conditions.

The stronger ambient IM3 hypothesis was not established. Four candidate parent pairs were freshly tested. Their product-only algebraic reconstruction errors were numerically modest, but the putative products had negative SNR and, in several cases, excessive baseband tracking error. The intermodulation graph and product-only reconstruction gates therefore failed. The final physical status is

$$\boxed{\text{RECEIVER_NONLINEARITY_PASS_AMBIENT_IM3_NOT_ESTABLISHED}}. \quad (12.1)$$

The central outcome is thus a disciplined partial validation. The receiver’s nonlinear state dependence is real, repeatable, and spectrally structured; the ambient data do not justify claiming causal hidden-parent recovery. This boundary is the most important conclusion of the work and provides a rigorous starting point for a later controlled two-tone experiment.

Complete Final V3 Survey Clipping Matrix

The following table records the clipping fraction, expressed as a percentage of complex samples satisfying the clipping indicator in eq. (3.10), for every final V3 survey center and gain state. These data make the safe/stress distinction visible in numeric form and preserve the complete clipping result of the final 28-center survey.

Table A.1: Final V3 clipping percentage at every survey center and gain state.

Center (MHz)	8.7 dB	14.4 dB	19.7 dB	29.7 dB	33.8 dB
88.720	0.000	0.000	0.000	0.000	0.000
89.420	0.000	0.000	0.000	0.000	0.000
90.120	0.000	0.000	0.000	1.630	43.700
90.820	0.000	0.000	0.000	0.610	60.180
91.520	0.000	0.000	0.000	12.730	34.140
92.220	0.000	0.000	0.000	0.520	11.130
92.920	0.000	0.000	0.000	0.350	12.020
93.620	0.000	0.000	0.000	0.000	0.190
94.320	0.000	0.000	0.000	33.220	50.470
95.020	0.000	0.000	0.000	45.160	63.220
95.720	0.000	0.000	0.340	42.140	50.960
96.420	0.000	0.000	0.000	7.780	17.410
97.120	0.000	0.000	0.000	13.450	23.000
97.820	0.000	0.000	0.000	25.640	39.800
98.520	0.000	0.000	0.000	27.230	43.180
99.220	0.000	0.000	0.000	3.730	9.400
99.920	0.000	0.000	0.000	0.440	8.320
100.620	0.000	0.000	0.000	0.380	11.030
101.320	0.000	0.000	0.000	0.000	0.000
102.020	0.000	0.000	0.000	0.000	0.090
102.720	0.000	0.000	0.000	0.000	0.290
103.420	0.000	0.000	0.000	0.000	8.430
104.120	0.000	0.000	0.000	0.000	7.430

Center (MHz)	8.7 dB	14.4 dB	19.7 dB	29.7 dB	33.8 dB
104.820	0.000	0.000	0.000	0.010	5.790
105.520	0.000	0.000	0.000	6.240	27.200
106.220	0.000	0.000	0.000	1.040	26.640
106.920	0.000	0.000	0.000	0.010	18.300
107.280	0.000	0.000	0.000	0.000	0.000

Complete Final V3 Target Gain Sweep

Table B.1: All final V3 target gain-sweep captures.

Gain (dB)	Repeat	Target power (dB)	SNR (dB)	Clip (%)
0.9	1	-28.822	21.78	0.000
0.9	2	-29.153	21.43	0.000
3.7	1	-23.465	26.09	0.000
3.7	2	-23.639	25.44	0.000
8.7	1	-20.218	28.04	0.000
8.7	2	-20.354	27.84	0.000
14.4	1	-15.608	29.64	0.000
14.4	2	-15.534	31.16	0.000
19.7	1	-7.701	33.12	0.000
19.7	2	-7.438	32.88	0.000
25.4	1	-2.724	26.66	9.520
25.4	2	-2.611	26.54	10.400
29.7	1	-1.040	22.55	34.730
29.7	2	-0.996	22.91	35.270
33.8	1	-0.430	21.62	53.320
33.8	2	-0.458	21.47	52.320
38.6	1	-0.084	21.22	65.670
38.6	2	-0.158	20.60	65.040
42.1	1	-0.120	20.79	66.710
42.1	2	-0.064	21.24	67.250
44.5	1	0.012	21.32	68.620
44.5	2	0.017	21.11	68.230
49.6	1	-0.026	20.83	68.790
49.6	2	-0.023	20.70	68.480

Complete Final V3 LO-Tracking Measurements

Table C.1: All six LO-coordinate validation captures of the final target.

LO (MHz)	Rep.	Exp. BB (kHz)	Det. BB (kHz)	Det. RF (MHz)	SNR (dB)
94.447007	1	650.000	653.434	95.100441	7.58
94.447007	2	650.000	653.604	95.100611	10.74
94.647007	1	450.000	453.873	95.100880	8.23
94.647007	2	450.000	454.120	95.101127	7.46
94.847007	1	250.000	254.576	95.101583	2.80
94.847007	2	250.000	253.117	95.100124	1.38

Complete Final Ambient IM3 Candidate Results

Table D.1: Final four ambient IM3 candidate pairs. Frequencies are in MHz, SNR values in dB, and reconstruction error in kHz.

Pair	f_1	f_2	$g_{1,p}$	$g_{2,p}$	$g_{1,o}$	$g_{2,o}$	SNR ₁	SNR ₂	Error	Gate
1	95.097007	98.494559	91.699455	101.892111	91.648581	101.898316	-20.43	-19.49	31.848	FAIL
2	95.489806	96.394546	94.585067	97.299285	94.520698	97.232392	-5.47	-20.41	66.051	FAIL
3	96.394546	98.197155	94.591937	99.999764	94.523461	99.998798	-6.46	-23.96	45.972	FAIL
4	95.097007	98.197155	91.996859	101.297302	92.075781	101.369122	-12.37	-21.64	76.554	FAIL

Campaign A and Campaign B Target-Sweep Records

Campaign A: Initial Real-World Engine

Gain (dB)	Repeat	Target power (dB)	SNR (dB)	Clip (%)
0.9	1	-35.406	16.56	0.000
0.9	2	-35.342	16.25	0.000
3.7	1	-29.448	21.50	0.000
3.7	2	-30.221	20.95	0.000
8.7	1	-25.431	24.64	0.000
8.7	2	-25.051	24.79	0.000
14.4	1	-20.100	27.96	0.000
14.4	2	-20.193	28.55	0.000
19.7	1	-11.971	30.36	0.000
19.7	2	-12.340	27.12	0.000
25.4	1	-6.967	31.01	0.014
25.4	2	-7.344	30.62	0.003
29.7	1	-3.741	27.32	7.633
29.7	2	-3.705	27.55	8.112
33.8	1	-1.978	23.31	26.982
33.8	2	-2.074	22.97	27.087
38.6	1	-1.291	20.10	49.601
38.6	2	-1.055	20.29	51.262
42.1	1	-1.089	19.48	55.878
42.1	2	-0.937	19.66	56.812
44.5	1	-0.977	19.08	58.963
44.5	2	-0.824	19.35	60.284
49.6	1	-1.125	18.31	59.904
49.6	2	-1.172	18.14	59.733

Campaign B: Rigorous V2

Gain (dB)	Repeat	Target power (dB)	SNR (dB)	Clip (%)
0.9	1	-36.606	14.69	0.000
0.9	2	-37.009	14.17	0.000
3.7	1	-30.667	19.89	0.000
3.7	2	-30.811	19.74	0.000
8.7	1	-26.606	23.09	0.000
8.7	2	-25.829	24.28	0.000
14.4	1	-21.332	27.55	0.000
14.4	2	-21.459	27.12	0.000
19.7	1	-15.028	30.51	0.000
19.7	2	-15.050	30.43	0.000
25.4	1	-9.856	30.45	0.000
25.4	2	-9.513	31.17	0.000
29.7	1	-5.377	31.65	0.000
29.7	2	-5.388	32.25	0.000
33.8	1	-1.511	30.51	7.480
33.8	2	-1.772	30.22	5.825
38.6	1	0.374	26.58	55.358
38.6	2	0.395	26.83	55.839
42.1	1	0.513	25.03	63.294
42.1	2	0.497	25.08	63.138
44.5	1	0.548	24.18	65.823
44.5	2	0.550	24.21	65.924
49.6	1	0.547	23.43	66.343
49.6	2	0.487	23.15	64.901

Final Gate Matrix and Interpretation

Table F.1: Final physical gate matrix.

Gate	Status
Asynchronous smoke capture	PASS
Survey coverage	PASS
Non-stale IQ	PASS
Real RF target	PASS
Nonlinear spectral redistribution	PASS
Receiver compression	PASS
Real RF LO tracking	PASS
Intermodulation graph	FAIL
Product-only parent reconstruction	FAIL

The corresponding summary is:

Hardware / measurement : PASS, (F.1)

Receiver nonlinearity : PASS, (F.2)

Ambient intermodulation graph : FAIL, (F.3)

Product-only reconstruction : FAIL. (F.4)

The status should be read as a layered conclusion, not as a contradiction. The first two statements concern phenomena that were directly and repeatedly observed. The last two concern a stronger causal interpretation that the ambient data did not support.

Reproducibility Record

Final campaign constants

Parameter	Final value
Receiver	RTL-SDR Blog V4, R828D tuner
Sample rate	2.048 MS/s
Complex samples per capture	250,000
Nominal capture duration	0.122070 s
Survey RF band	88–108 MHz
Survey center count	28
Nominal center spacing	0.700 MHz
Final upper-edge center	107.280 MHz
Survey gains	8.7, 14.4, 19.7, 29.7, 33.8 dB
Safe gains	8.7, 14.4, 19.7 dB
Stress gains	29.7, 33.8 dB
FFT length	8192
FFT-bin spacing	250 Hz
RF stitching grid	10 kHz
Usable half-band per center	780 kHz
DC guard	55 kHz
Target half-band	approximately 110 kHz
Final selected target	95.097007 MHz
Robust residual scale	5.605 dB
Adaptive residual threshold	25.221 dB

Acquisition description

No source code is reproduced in this thesis. The final acquisition procedure can be summarized unambiguously in prose. For each requested center frequency and manual gain state, the V4-compatible RTL-SDR command-line receiver was configured for device 0, the requested

center frequency, a 2.048 MS/s complex sample rate, a nonzero manual tuner-gain command, and 250,000 complex samples. The output was written as interleaved unsigned 8-bit IQ bytes. Captures were accepted only when the expected 500,000-byte file was present and no fatal transfer condition was indicated. Survey gain order was randomized at each RF center. Spectral estimation and stitching were performed only after acquisition completed.

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